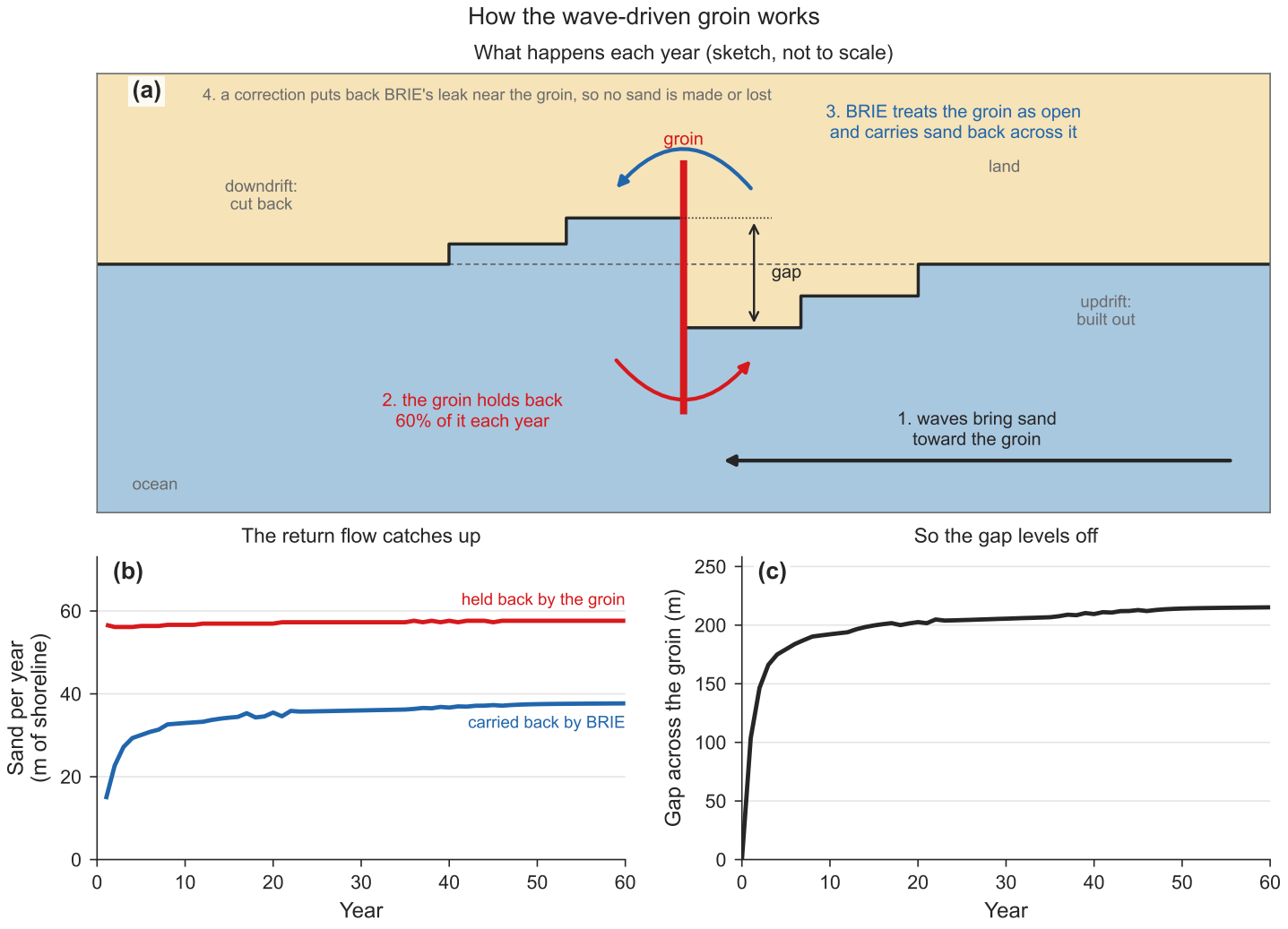
How the wave-driven groin works
What happens each year (sketch, not to scale)
(a)
4. a correction puts back BRIE's leak near the groin, so no sand is made or lost
3. BRIE treats the groin as open and carries sand back across it
groin
land
downdrift: cut back
gap
updrift: built out
2. the groin holds back 60% of it each year
1. waves bring sand toward the groin
ocean
The return flow catches up
(b)
held back by the groin
carried back by BRIE
Sand per year (m of shoreline)
0
20
40
60
0
10
20
30
40
50
60
Year
So the gap levels off
(c)
Gap across the groin (m)
0
50
100
150
200
250
0
10
20
30
40
50
60
Year